

CS 141 - Lab 4: Structs in C

Lab Objectives

By the end of this lab, you should be able to:

- Understand why we use `struct` in C
- Declare a `struct` type and create variables from it
- Access and modify struct fields using the dot operator (`.`)
- Read and print struct data using functions
- Use arrays of structs to store multiple records

Exercise 1: First Struct

1. Create `exercise1.c`
2. Define a `struct Student` with fields:
 - `char name[50]`
 - `int age`
 - `float grade`
3. In `main`, create one student variable.
4. Fill it with sample values.
5. Print all fields in a clear format.

Expected Output Example

```
Name: Alice
Age: 20
Grade: 16.50
```

Exercise 2: Reading Struct Data from User

1. Create `exercise2.c`
2. Reuse the `Student` struct.
3. Read name, age, and grade from user input.
4. Print a summary sentence, for example:
 - `Student Bob (age 19) has grade 14.25`

Hints

- Use `scanf("%49s", s.name)` for the name (single word).
- Use `%.2f` when printing grades.

Exercise 3: Array of Structs

1. Create `exercise3.c`
2. Declare an array of 5 students.
3. Use a loop to read data for each student.
4. Use another loop to print all students.
5. Compute and print:
 - Class average grade

Expected Output Example

```
Student 1: Sara, age 18, grade 15.00
Student 2: Omar, age 19, grade 13.50
...
Class average: 14.20
```

Exercise 4: Struct with Functions

1. Create `exercise4.c`
2. Define:
 - `void printStudent(struct Student s)`
 - `int isPassed(struct Student s)` (returns 1 if grade ≥ 10 , else 0)
3. In `main`, create one student and call both functions.
4. Print whether the student passed or failed.

Note: Passing struct by value copies it. This is fine for small structs in this lab.

Challenge Exercise

Objective: Build a mini student record manager.

1. Create `challenge.c`
2. Store 6 students in an array of structs.
3. Display a menu:
 1. Show all students
 2. Show highest grade student
 3. Show lowest grade student
 4. Search student by name
 5. Exit
4. Use loops + conditionals to implement each option.

Quick Reference

Define struct:

```
struct Student {  
    char name[50];  
    int age;  
    float grade;  
};
```

Create variable:

```
struct Student s;
```

Access field:

```
s.age = 20;  
printf("%d", s.age);
```